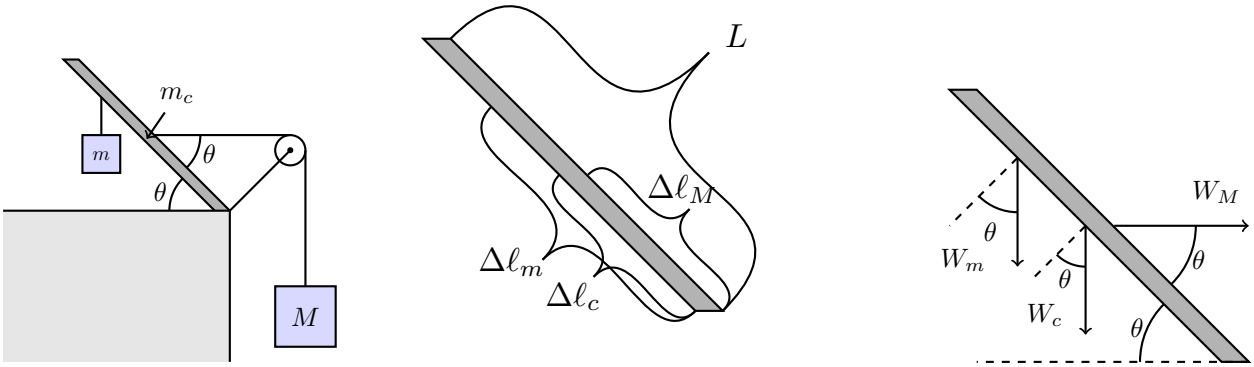
Physics 1101: Cable Stayed Design

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1 Warm-Up



We note that the main concern of this problem is rotational stability. This is an important note, because it makes the problem solving process much much easier! Because of this, the first principle we should use is setting the net torque equal to 0. Torque gives us our rotational stability, whereas we would use net *force* for positional stability.

Let's first just find the torques for each of the forces acting on our system. I will skip over the free body diagrams for each of the strings holding weights, because we know that a free-hanging weight on a string implies that the tension of the hanging string is just equal to the hanging weight.

$$\tau_m = \Delta\ell_m mg\cos\theta \tag{1}$$

$$\tau_c = \frac{L}{2}m_c g\cos\theta \tag{2}$$

Where m_c is the mass of the channel itself

$$\tau_M = -\Delta\ell_M Mg\sin\theta \tag{3}$$

$$\tag{4}$$

Thus the net torque is

$$\tau_{NET} = \Delta\ell_m mg\cos\theta + \frac{L}{2}m_c g\cos\theta - \Delta\ell_M Mg\sin\theta = 0 \tag{5}$$

We can solve this for the relation between the angle of the channel θ and the counterweight mass m.

$$\Delta\ell_M M g \sin\theta = \Delta\ell_m m g \cos\theta + \frac{L}{2} m_c g \cos\theta \tag{6}$$

$$\Delta\ell_M M \sin\theta = \Delta\ell_m m \cos\theta + \frac{L}{2} m_c \cos\theta \tag{7}$$

Dividing both sides by cos

$$\Delta\ell_M M \frac{\sin\theta}{\cos\theta} = \Delta\ell_m m \cos\theta + \frac{L}{2} m_c \cos\theta \tag{8}$$

$$\Delta\ell_M M \tan\theta = \Delta\ell_m m \cos\theta + \frac{L}{2} m_c \cos\theta \tag{9}$$

$$\tan\theta = \frac{\Delta\ell_m m + \frac{L}{2} m_c}{\Delta\ell_M M} \tag{10}$$

$$\tan\theta = \frac{\Delta\ell_m m}{\Delta\ell_M M} + \frac{\frac{L}{2} m_c}{\Delta\ell_M M} \tag{11}$$

$$\theta = \arctan\left(\frac{\Delta\ell_m}{\Delta\ell_M M} m + \frac{\frac{L}{2} m_c}{\Delta\ell_M M}\right) \tag{12}$$

Where arctan (also called $\tan^{-1}$) is the inverse tangent function.

